Michael Huang
EN.555.644.81
Homework 4

5.2.

What is the difference between the forward price and the value of a forward contract?

The forward price is the actual price at which the asset would be agreed to be bought or sold at the set future time, while the value of the actual contract is the actual positive or negative return that the contract would generate as the spot price changes over time, starting initially at 0. So one is a specification of the contract while the other is a result from the specification of that contract.

5.4.

A stock index currently stands at 350. The risk-free interest rate is 8% per annum (with continuous compounding) and the dividend yield on the index is 4% per annum. What should the futures price for a four-month contract be?

We have an asset with a known yield, so we solve using

$$F_0 = S_0 e^{(r-q)T}$$

$$= 350 \cdot e^{(0.08-0.04)\cdot\frac{4}{12}}$$

$$= 350 \cdot e^{0.04\cdot\frac{1}{3}}$$

$$= 350 \cdot 1.0134226186$$

$$F_0 = 354.69791651$$

so the futures price is $\boxed{\textbf{\$354.70}}$.

5.6

Explain carefully the meaning of the terms convenience yield and cost of carry. What is the relationship between futures price, spot price, convenience yield, and cost of carry?

The convenience yield essentially measures the return or level of benefit of directly holding on to a physical asset rather than owning futures contract on the physical asset. The cost of carry is the net cost of holding, i.e. carrying, the actual physical asset, which we define as the sum of the storage cost and the interest cost of financing the acquisition of the held asset.

To define the relationship between futures price, spot price, convenience yield, and cost of carry, we define the following equation:

$$F_0 = S_0 e^{(c-y)T}$$

with futures price F_0, spot price S_0, convenience yield y, cost of carry c, and time to the corresponding futures contract maturity T. We calculate the futures price accordingly, given those other variables.

5.7

Explain why a foreign currency can be treated as an asset providing a known yield.

A foreign currency can be treated as an asset providing a known yield – i.e. at a predictable rate relative to the asset's actual value – because the foreign currency provides some level of interest by just holding it. This interest accumulated by holding the foreign currency is a known percentage relative to the value of the foreign currency, so we can therefore predictably estimate the yield from the asset.

5.12

Suppose that the risk-free interest rate is 10% per annum with continuous compounding and that the dividend yield on a stock index is 4% per annum. The index is standing at 400, and the futures price for a contract deliverable in four months is 405. What arbitrage opportunities does this create?

We can calculate the following futures price:

$$F_0 = S_0 e^{(c-y)T}$$

$$= 400 \cdot e^{(0.10-0.04)\cdot\frac{4}{12}}$$

$$= 400 \cdot e^{0.02}$$

$$F_0 = 408.080536012$$

This theoretical futures price is higher than the actual futures price for the contract at 405. This indicates that the futures price is too low relative to the index, so to arbitrage, we should **buy the futures contracts that are cheap relative to its actual value, and then short the underlying stocks in the index, investing the returns from the short sale to invest at the risk-free rate.** At maturity, we would pay the dividends to the lender, then take delivery via the futures contract at 405 and use that to return the borrowed index and close the short position.

5.16

Suppose that F_1 and F_2 are two futures contracts on the same commodity with times to maturity, t_1 and t_2, where $t_2 > t_1$. Prove that

$$F_2 \leq F_1 e^{r(t_2-t_1)}$$

where r is the interest rate (assumed constant) and there are no storage costs. For the purposes of this problem, assume that a futures contract is the same as a forward contract.

Assuming that a futures contract is the same as a forward contract here, we are essentially saying that the price of the future with later maturity date is less than or equal to the price of the earlier maturing futures contract at its maturity time t_1 multiplied by $e^{r(t_2-t_1)}$, which we know to be equivalent to the continuously compounded interest at rate r from t_1 to t_2. We can easily see that this term is equal to the amount of money it costs to buy the commodity at t_1 – F_1 – plus the interest on the money it cost buy that commodity, $F_1e^{r(t_2-t_1)} - F_1$.

Using this information, we can test the inequality by seeing if it is possible that $F_2 > F_1e^{r(t_2-t_1)}$, that is, $F_2 - F_1e^{r(t_2-t_1)} > 0$. Looking at this, we set up the following situation: we buy the underlying commodity at t_1 with a loan until t_2, which represents the second term, i.e. we short the future with maturity t_1. The first term therefore just represents a long on the future with maturity t_2. So therefore there must not be an arbitrage opportunity when we long F_2 and short F_1. However, in reality, if this risk-free arbitrage opportunity existed, then the two terms should eventually equalize, so this situation of $F_2 - F_1e^{r(t_2-t_1)} > 0$ cannot exist, which essentially shows that $F_2 - F_1e^{r(t_2-t_1)} \leq 0$ or $F_2 \leq F_1e^{r(t_2-t_1)}$, as we initially sought to prove.

5.17.

When a known future cash outflow in a foreign currency is hedged by a company using a forward contract, there is no foreign exchange risk. When it is hedged using futures contracts, the daily settlement process does leave the company exposed to some risk. Explain the nature of this risk. In particular, consider whether the company is better off using a futures contract or a forward contract when...

Assume that the forward price equals the futures price.

a)

The value of the foreign currency falls rapidly during the life of the contract

When the value of the foreign currency falls rapidly during the life of the contract, the company will lose money regardless, but with the forward contract, this happens all at the end of the contract while with the futures contract the loss will be realized daily early during the duration of the contract, so we start paying margin immediately, which is worse on a present value basis since we lose out on gains on interest immediately. So we $\boxed{\textbf{prefer the forward contract}}$ in this case.

b)

The value of the foreign currency rises rapidly during the life of the contract

When the value of the foreign currency rises rapidly during the life of the contract, the company realizes gains daily when using the futures contract rather than all at the end with the forward contract, so on a present value basis, we would $\boxed{\textbf{prefer the futures contract}}$.

c)

The value of the foreign currency first rises and then falls back to its initial value

In this case, the outcome at maturity is the same, with no overall loss or gain with either contract. On a present value basis, with futures contracts, the company first realizes some gains daily when the foreign currency is rising, receiving positive cash flows and interest first before paying the original cash back. So in this case, we would $\boxed{\textbf{prefer the futures contract}}$.

d)

The value of the foreign currency first falls and then rises back to its initial value

In this case, the outcome at maturity is also the same, with no overall loss or gain. On a present value, with the futures contract, the company first has to pay margin on the losses, and then is reimbursed. This initial payment and loss is not preferable to just taking 0 overall PnL with the forward contract, so in this case we $\boxed{\textbf{prefer the forward contract}}$.

5.20

Show that equation (5.3) is true by considering an investment in the asset combined with a short position in a futures contract. Assume that all income from the asset is reinvested in the asset. Use an argument similar to that in footnotes 2 and 4 of this chapter and explain in detail what an arbitrageur would do if equation (5.3) did not hold.

We aim to derive equation (5.3), i.e. that

$$F_0 = S_0 e^{(r-q)T}$$

Using the same formatting from the text, where we know that if all income is reinvested in the asset, then the number of units held of the asset grows at rate q during time T, where q is the average yield on the asset within the time period. We want to set up a situation where we buy exactly enough at time 0 to to deliver one unit of the asset and receive F_0, the price of the asset at T, thus closing the short futures contract. This amount we buy x becomes the same value as one unit of the asset when reinvested, i.e.

$$1 = x \cdot e^{qT}$$
$$x = \frac{1}{e^{qT}}$$
$$x = e^{-qT}$$

So we invest in e^{-qT} units of the asset, at the total cost of $S_0 \cdot e^{-qT}$. We also need to enter this into a short futures contract, to sell the one unit of the asset which we will eventually hold at T for the F_0 which we are due to receive. We know that we are guaranteed to receive F_0, so this is a risk-free investment which we discount accordingly at the risk-free rate r, so we end up discounting to get to the present value of $F_0 e^{-rT}$ with the setup cost of $S_0 e^{-qT}$, i.e.

$$F_0 e^{-rT} = S_0 e^{-qT}$$

$$F_0 = S_0 e^{(r-q)T}$$

exactly as we sought to show.

If this equation did not hold, the arbitrageur would be able to profit in the following ways:

- If $F_0 > S_0e^{(r-q)T}$, i.e. the price of the forward is higher than the investment cost, then the arbitrageur should borrow S_0e^{-qT} at interest rate r, and invest this in e^{-qT} units of the asset, which as we showed above eventually reaches the value of F_0, or 1 asset. At the same time, the arbitrageur enters a short futures contract to sell 1 unit of the asset at time T for the price of F_0. At T, the arbitrageur then therefore has 1 unit of the actual asset, which they use to fulfill the short futures contract and deliver the unit, receiving F_0. Finally, they then pay off the loan amount which has grown to $S_0e^{(r-q)T}$, which leads to profit in this case since our total cash flow is $F_0 - S_0e^{(r-q)T}$, which in our first assumption $F_0 > S_0e^{(r-q)T}$ indicates that this is positive, and therefore a profitable strategy.
- If $F_0 < S_0e^{(r-q)T}$, i.e. the price of the forward is lower than the investment cost, then the arbitrageur should do the reverse – they should sell e^{-qT} of the asset to instantly get S_0e^{-qT} which they should then invest at the risk-free rate r. Then, the arbitrageur should enter into a long forward contract to buy 1 unit of the asset at time T for F_0. The money gained from initially buying the asset has now grown at the risk free rate to $S_0e^{(r-q)T}$, and they can then buy the 1 unit of the asset for F_0. This newly acquired $1 - e^{-qT}$ of the asset is the same as it would have been had we just reinvested the returns of the e^{-qT} we initially sold, so we come out with 1 unit of asset just as we would have without running this strategy, except with $S_0e^{(r-q)T} - F_0$ cash left over, which is a positive amount, according to our initial assumption $F_0 < S_0e^{(r-q)T}$.

5.27

An index is 1,200. The three-month risk-free rate is 3% per annum and the dividend yield over the next three months is 1.2% per annum. The six-month risk-free rate is

3.5% per annum and the dividend yield over the next six months is 1% per annum. Estimate the futures price of the index for three-month and six-month contracts. All interest rates and dividend yields are continuously compounded.

For the three-month contract:

$$F_0 = S_0 e^{(c-y)T}$$

$$= 1200 \cdot e^{(0.03-0.012)\cdot\frac{3}{12}}$$

$$= 1200 \cdot e^{0.018\cdot\frac{1}{4}}$$

$$= 1200 \cdot 1.0045101402$$

$$F_0 = 1205.41216825$$

So the futures price of the index for the three-month contract is approximately $\boxed{\textbf{\$1,205.41}}$. For the six-month contract:

$$F_0 = S_0 e^{(c-y)T}$$

$$= 1200 \cdot e^{(0.035-0.01)\cdot\frac{6}{12}}$$

$$= 1200 \cdot e^{0.025\cdot\frac{1}{2}}$$

$$= 1200 \cdot 1.01257845154$$

$$F_0 = 1215.09414185$$

So the futures price of the index for the six-month contract is approximately $\boxed{\textbf{\$1,215.09}}$.